



Time Series
Analysis

Time Series Analysis

Today's Topic: Multivariate Time Series II



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<https://sithuaung-ink.github.io/data-analyst-portfolio/>

Vector Autoregressive Moving Average (VARMA) model

A VARMA(p, q) model

$$\mathbf{X}_t = \boldsymbol{\mu} + \sum_{i=1}^p \boldsymbol{\Phi}_i \mathbf{X}_{t-i} + \sum_{j=1}^q \boldsymbol{\Theta}_j \boldsymbol{\varepsilon}_{t-j} + \boldsymbol{\varepsilon}_t$$

where $\boldsymbol{\mu}$ is mean vector, $\boldsymbol{\Phi}_i$ is AR matrices, $\boldsymbol{\Theta}_j$ is MA matrices, and $\boldsymbol{\varepsilon}_t$ is a white noise.

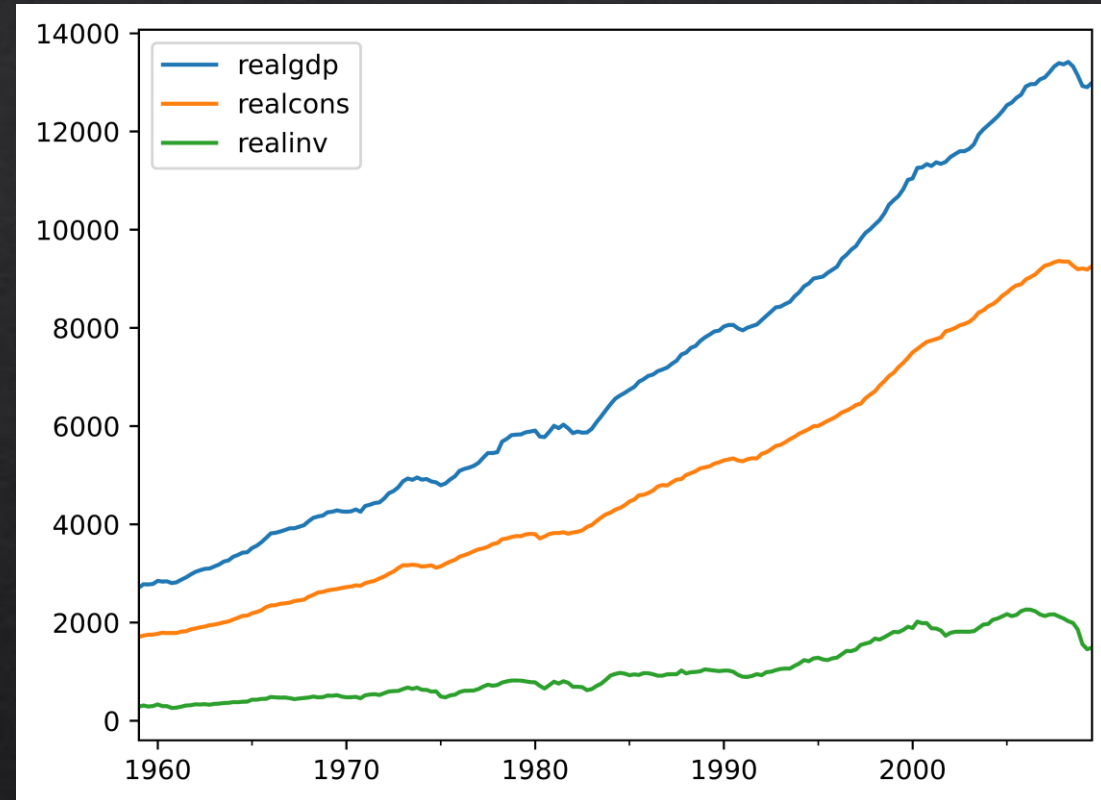
K -dimensional vector time series $\mathbf{X}_t = (\mathbf{X}_{t1}, \dots, \mathbf{X}_{tK})'$ where $\mathbf{X}_{tk}$ represent the return at time t of $\mathbf{X}_k$ ($1 \leq k \leq K$).

United States Macroeconomic Data

- Loads the US macroeconomic Dataset from *statsmodels* using `sm.datasets.macrodta.load_pandas().data`.
- The data is quarterly data from 1959Q1 to 2009 Q3.
- Total of 14 variables and 203 observations, but we selected real GDP (realgdp), real consumption (realcons), and real investment (realinv).

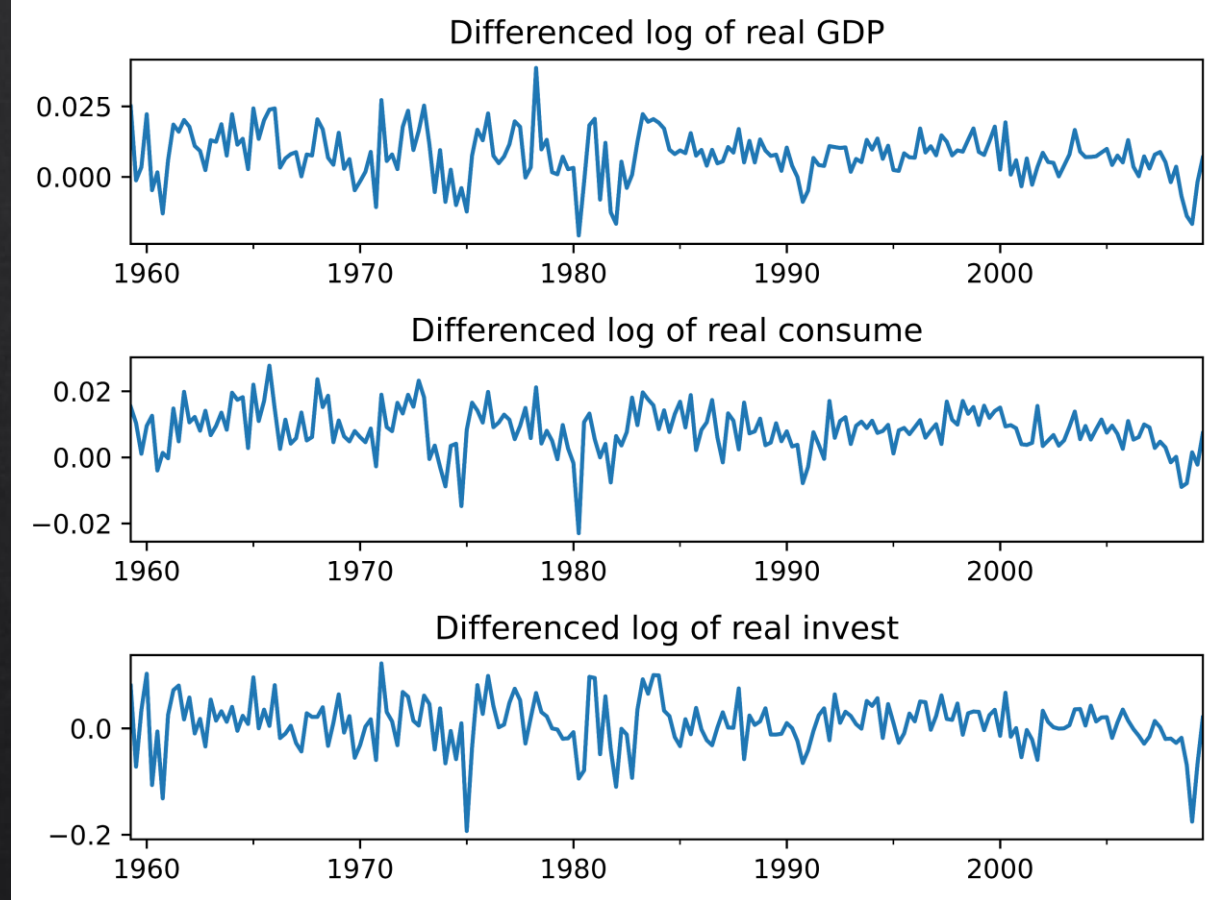
Time Series Plot

- The time series is a linear increasing trend.
- Taking a logarithm on the series and then first-differencing.

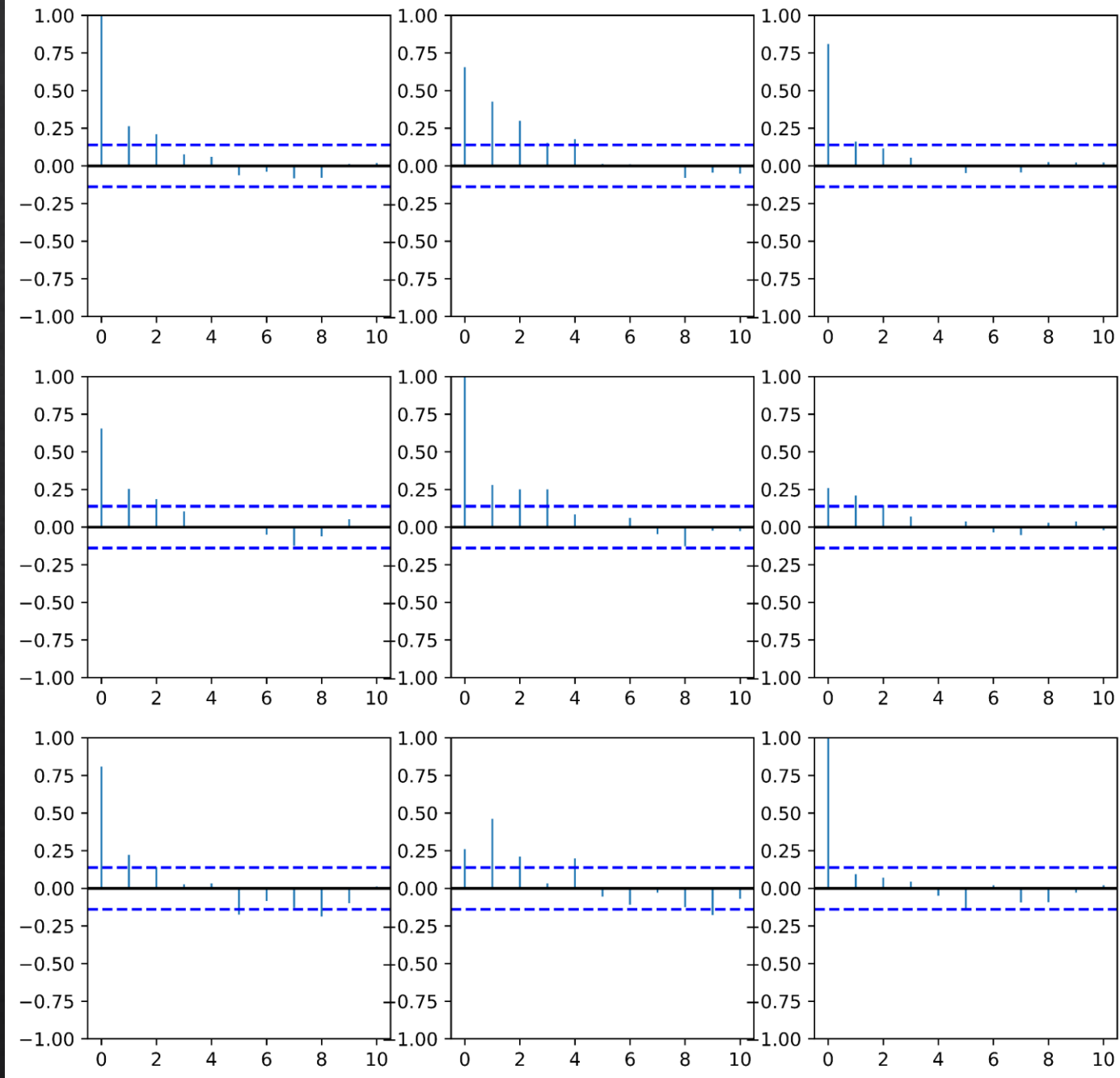


First Difference of Logarithms

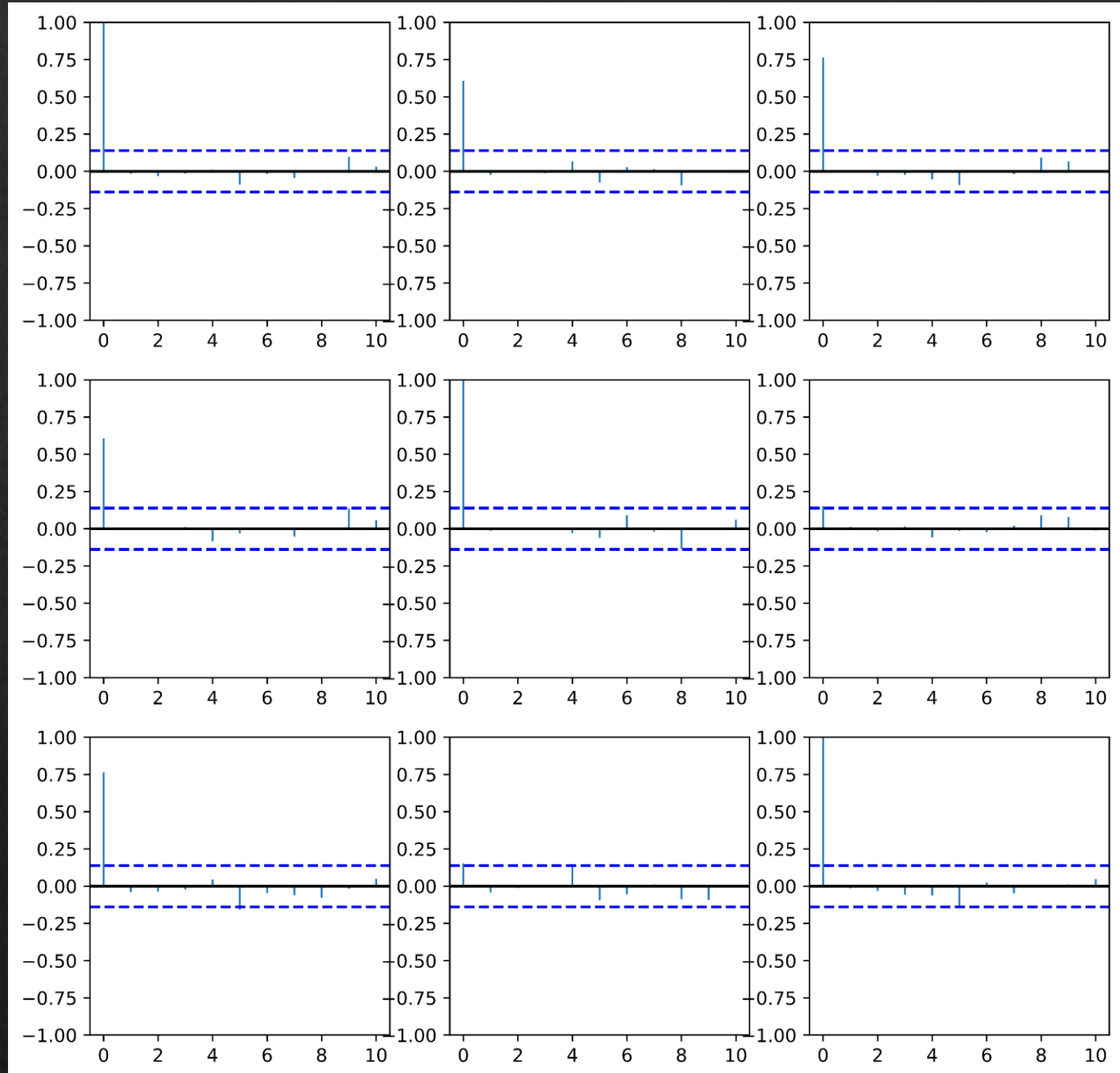
- The time series is now stationary.



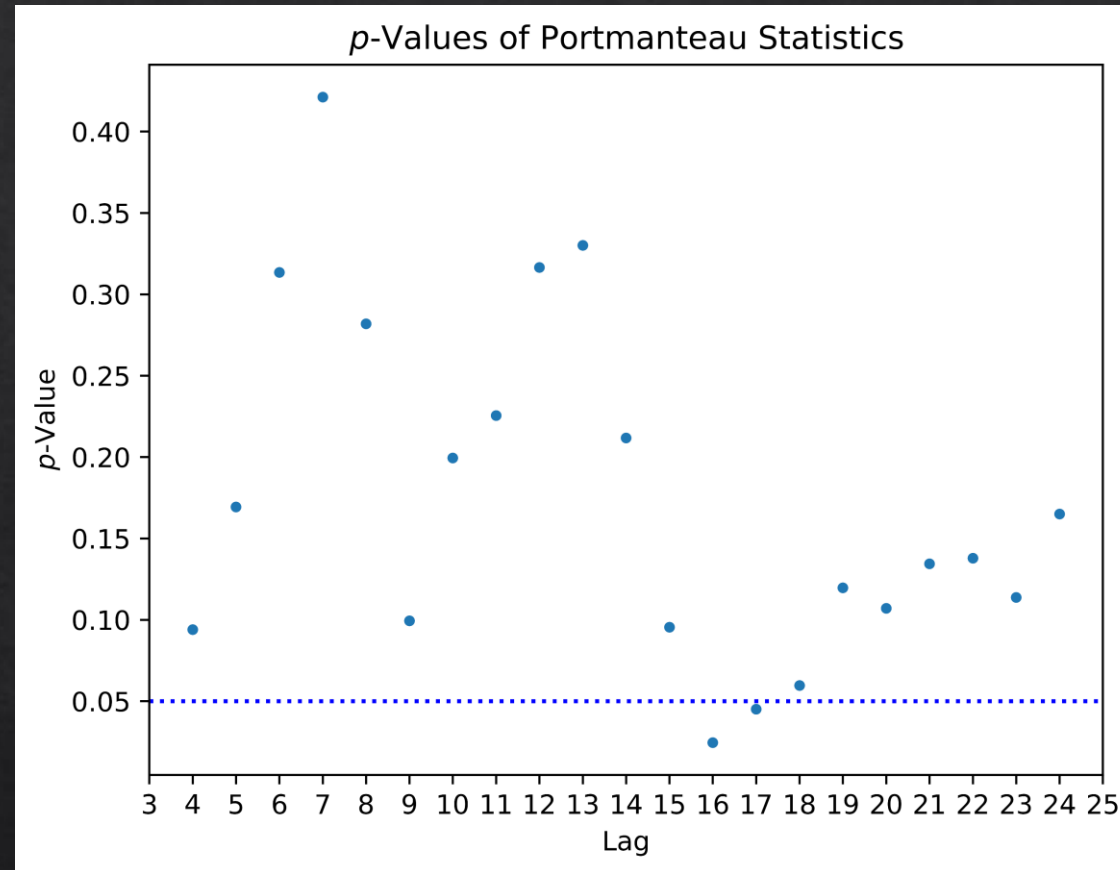
Correlation Plot of first differences of logs



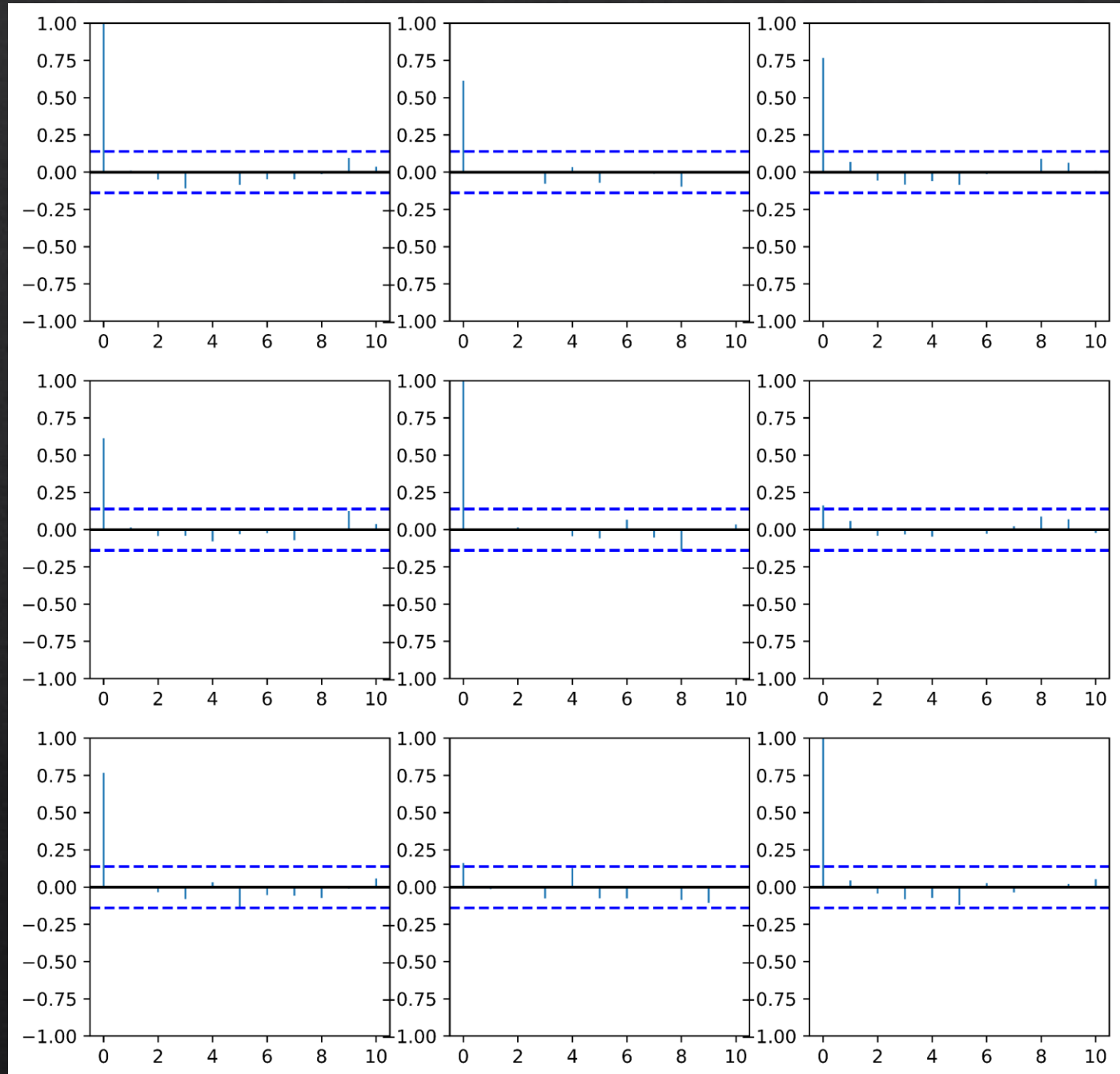
Correlation Plot of the VAR (3) model residual



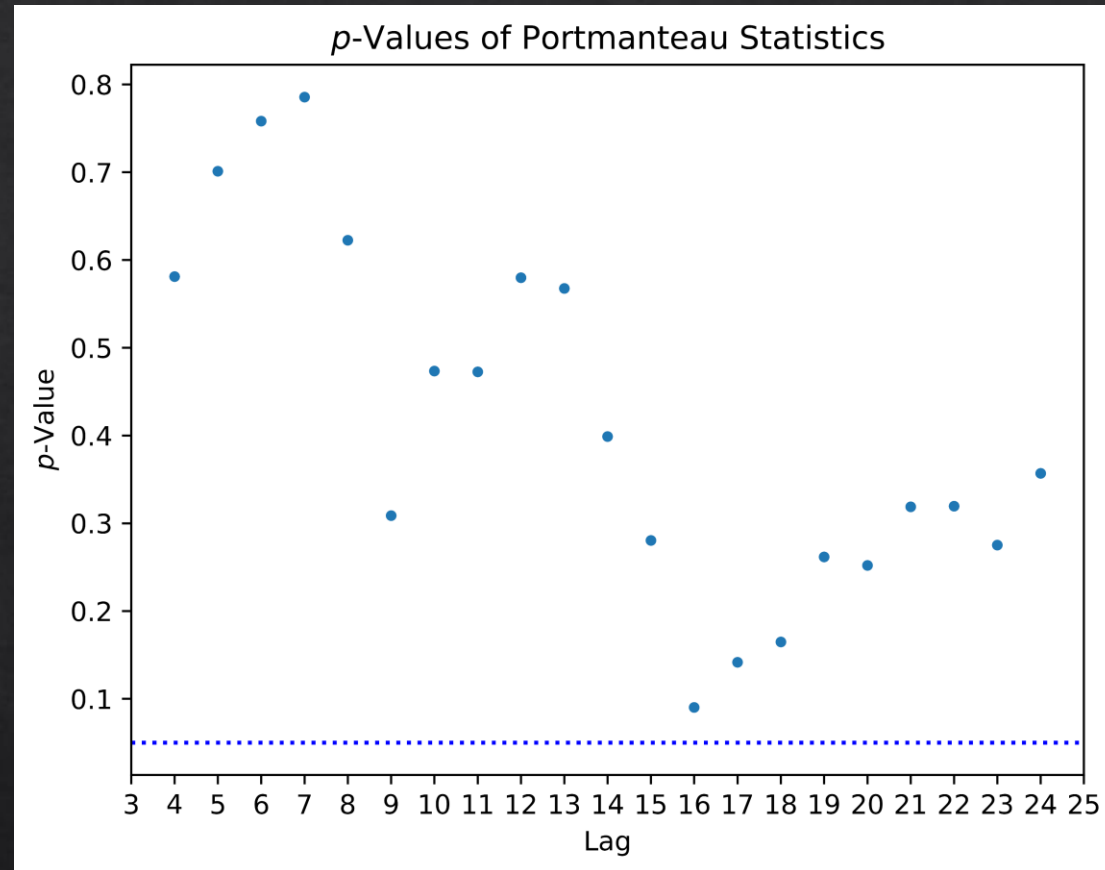
p -values plot of the model residual



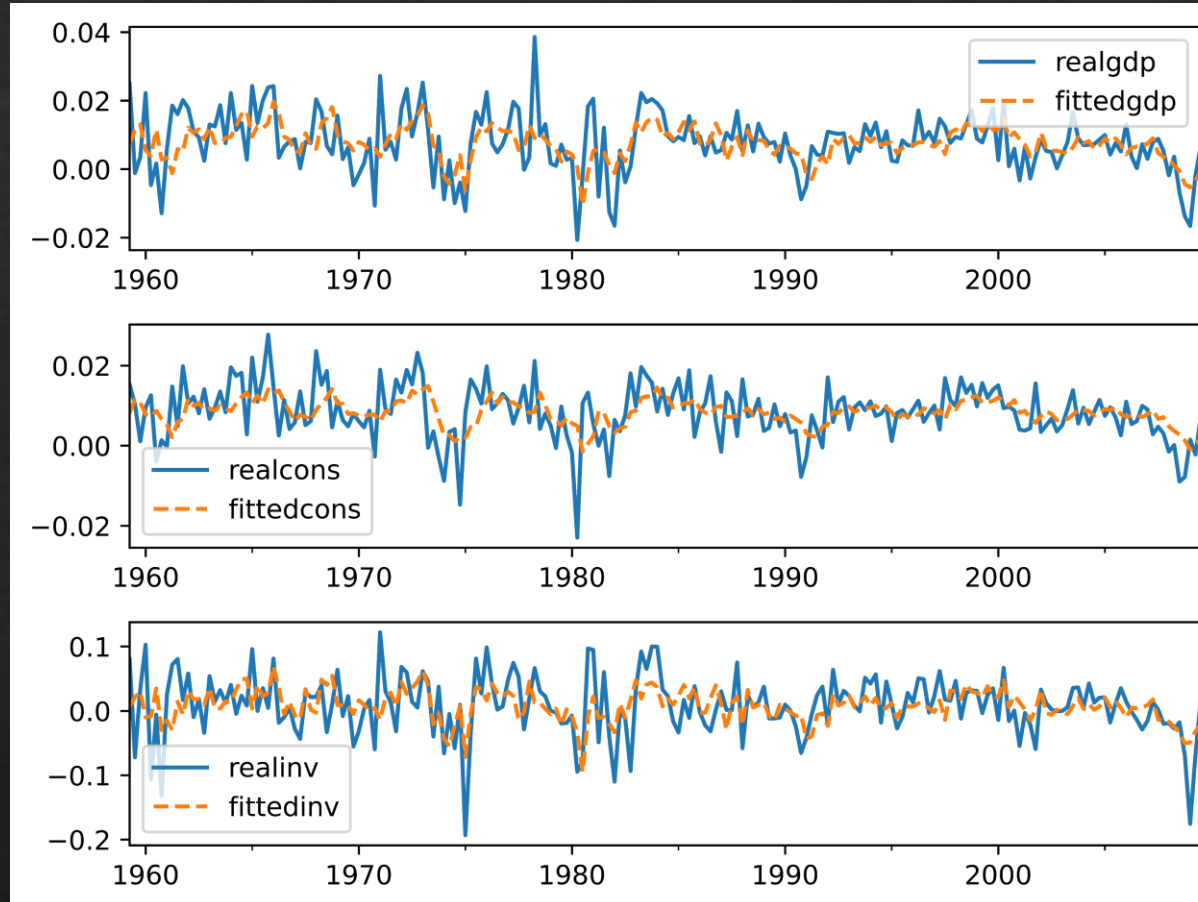
Correlation Plot of the refined model residual



p -values plot of the refined model residual



Predicting the future values with VAR(3) model



VARMA Model

- The residual of the initial VAR(3) model did not pass the p-value test, but the refined model passed the test.
- The model predictions are satisfactory in general, but the actual values are at spikes, so the fitted values are not so good.